

Native Plots with pgfplots

Solving and plotting a quadratic

Consider the function $f(x) = x^2 - 2x - 3$. Factoring gives:

$$\begin{aligned} f(x) &= x^2 - 2x - 3 \\ &= (x - 3)(x + 1) \end{aligned}$$

Setting $f(x) = 0$ shows that this function has two roots, at $x = -1$ and $x = 3$.

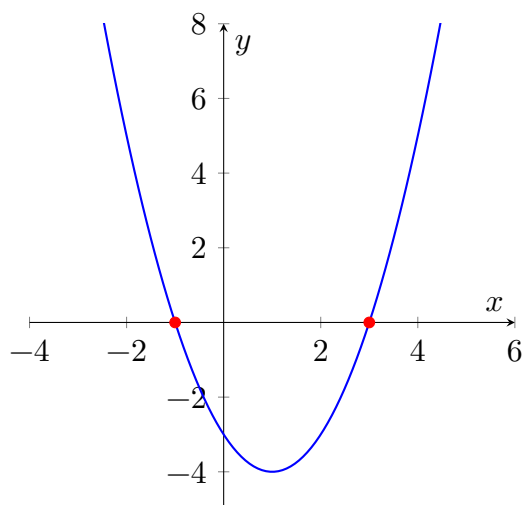


Figure 1: The graph of $f(x) = x^2 - 2x - 3$ crossing the x -axis at its two roots